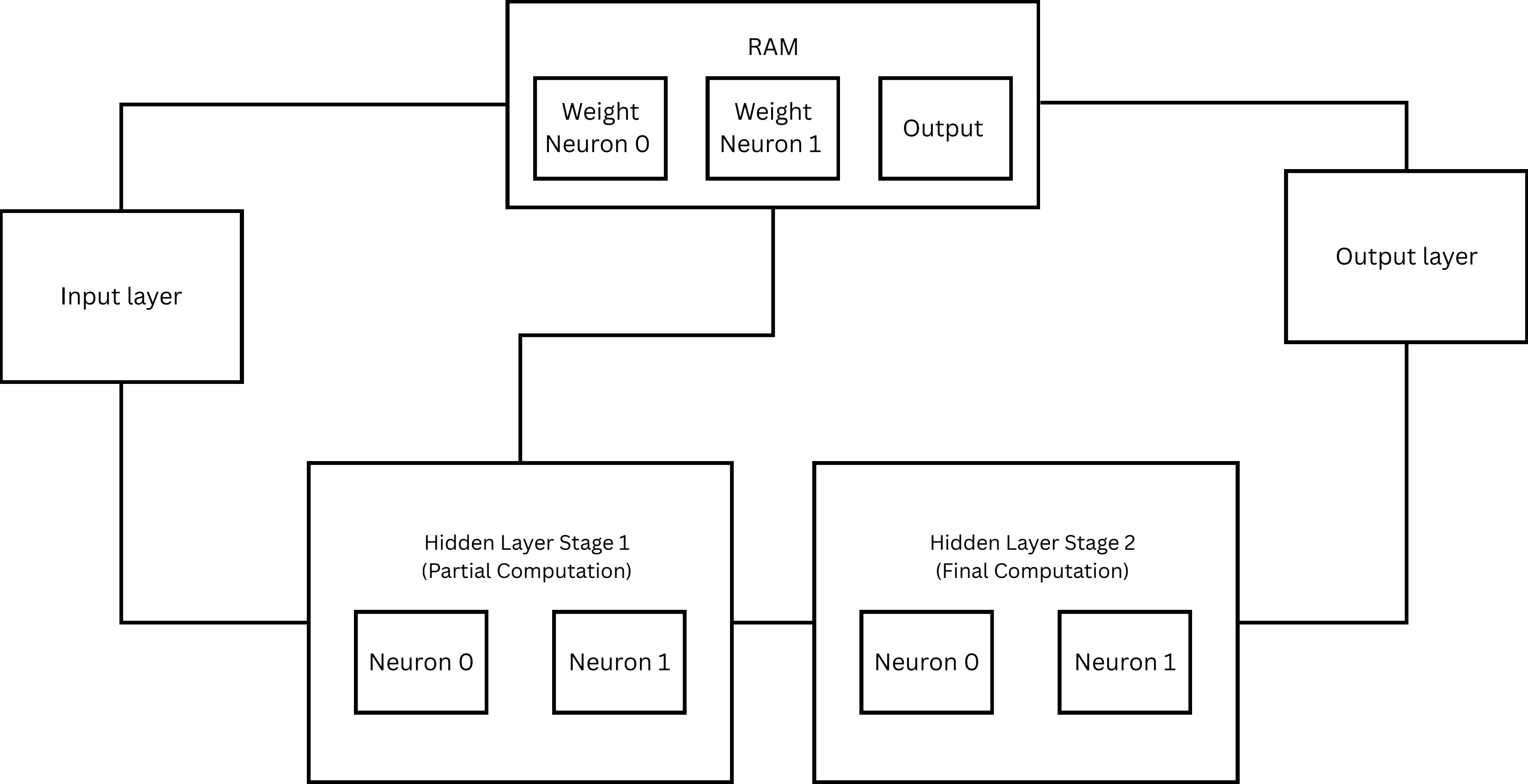
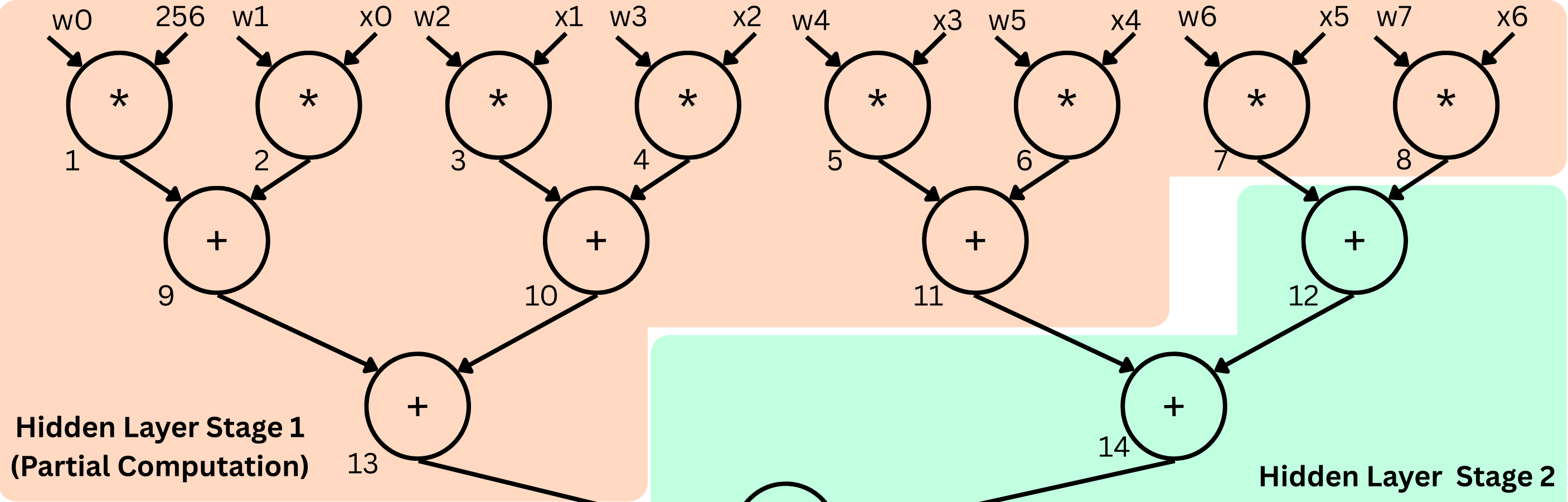


HDL Structural Diagram



HDL Optimizations

cycles	operations
0	1
1	2
2	3, 9
3	4
4	5, 10
5	6, 13
6	7, 11
7	8
8	12
9	14
10	15, 16

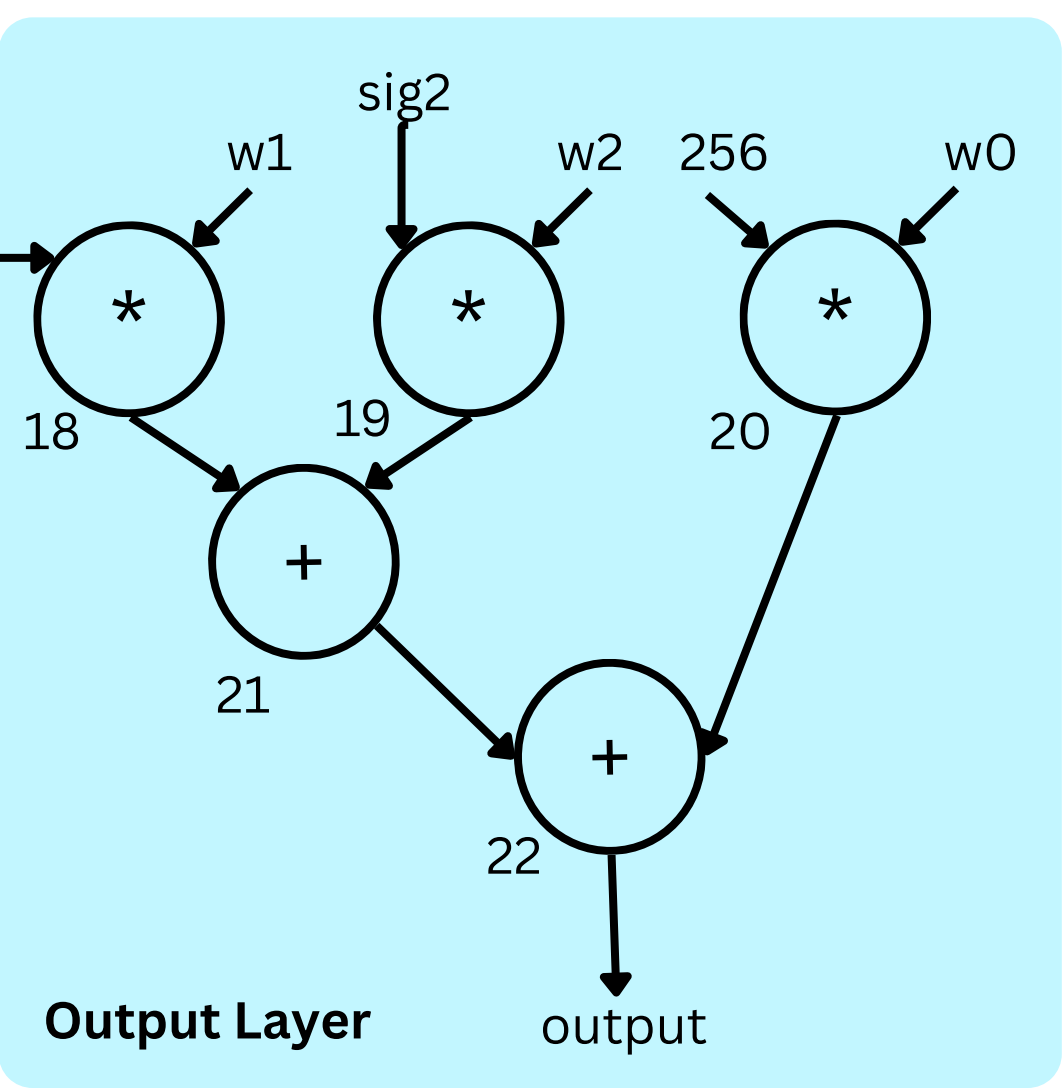


1. After cycle 7, Stage 1 begins processing the next input while Stage 2 continues computation of the previous input

2. **Computation overlaps** across Stage 1, Stage 2, and Output Layer, ensuring continuous data processing

3. Multiple stages operate concurrently, achieving **pipeline parallelism**

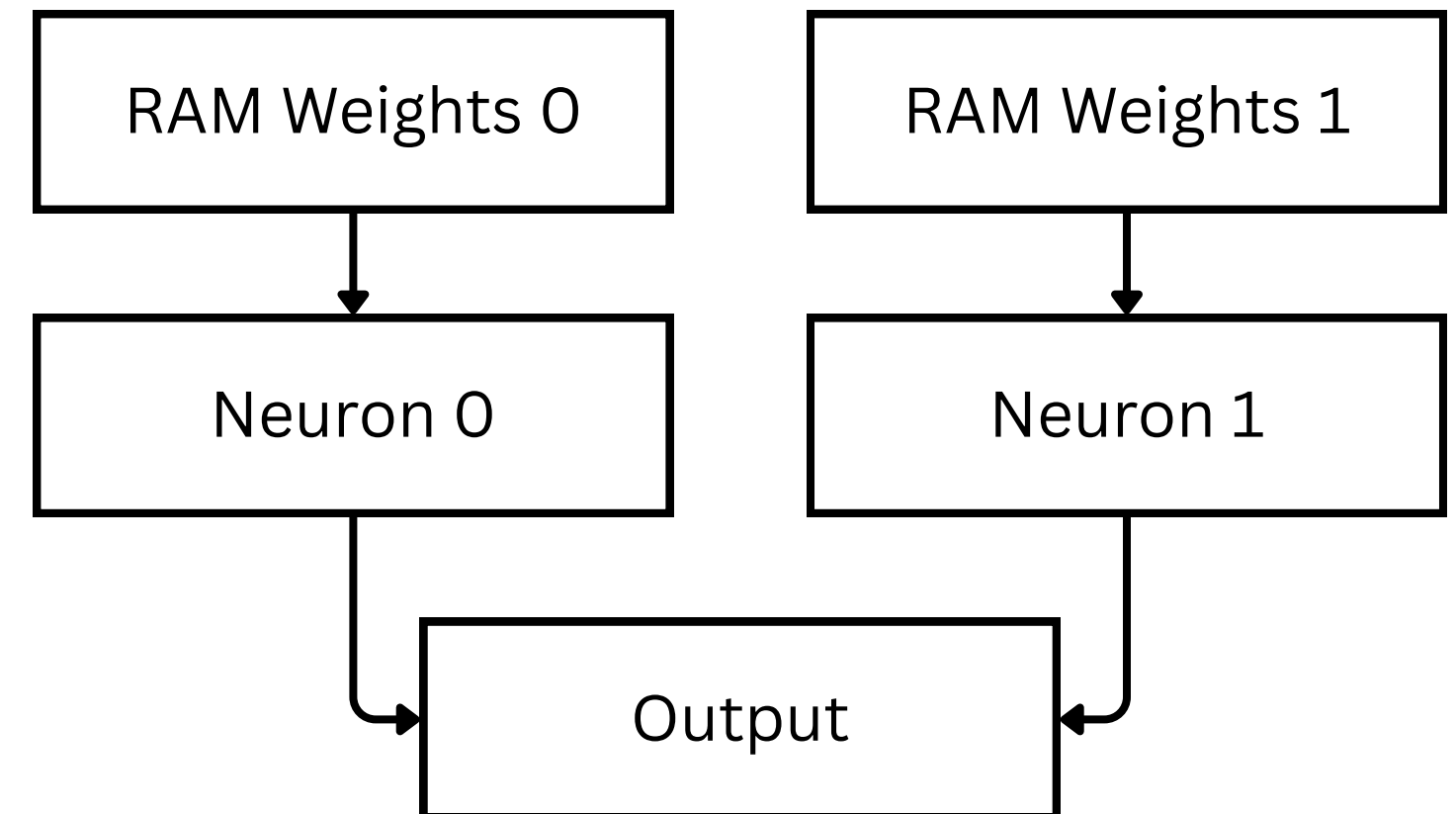
cycles	operations
11	17
12	18, 19, 20
13	21, 22



HDL Memory Architecture

- To support parallel neuron computation, we partition the weights across multiple memory modules.
- **Parallel Memory Access:** Each Neuron has its own dedicated memory (RAM Weights 0 and RAM Weights 1)
- **Parallel Computation:** Neurons compute concurrently without waiting for shared memory access

** The pipelined and partitioned design enables **streaming input processing**, eliminating the need for intermediate storage and avoiding additional memory access delays



TARGET	ESTIMATED	UNCERTAINTY
10.00 ns	7.075 ns	2.70 ns

Meets timing requirement of 10ns.

Resource Usage Comparison			
Resource	HDL	HLS	Difference
LUT	958	2,353	~2.4x increase
FF	762	1,795	~2.4x increase
BRAM/DSP	0/0	1/9	HLS optimization

Compute Time (In cycles/ticks)		
Soft	Hard HDL	Hard HLS
4718	1,125	887

HLS Optimizations

- Task-level pipelining (Dataflow)
- Stream-based FIFO interconnects for high throughput (Allows producer and consumer to work in parallel)
- Complete Array Partitioning

